

A Practical Introduction to Generative AI for Bioinformatics

NCI Bioinformatics Training and Education Program

2026-09-23

Today's Agenda

- ▶ What Is Generative AI?
- ▶ AI Tools at NIH
- ▶ Chat vs. Agentic AI
- ▶ AI in Bioinformatics Practice
- ▶ Getting Started

What Is Generative AI?

The AI Landscape: Sorting Out the Jargon

Note

A common source of confusion: People mix up companies, products, and models. These are not the same thing.

Level	Example	What it is
Company	Anthropic	Builds and maintains the AI
Product / Interface	Claude.ai	The app or website you use
Model	Claude Sonnet 4.5	The specific AI “brain” running
API	Anthropic API	How developers access the model

Think of it like: Ford (company) → Ford F-150 (product) → 5.0L V8 (engine/model)

What Makes AI “Generative”?

Generative AI creates new content including text, code, images, data rather than just classifying or retrieving existing content.

- ▶ Traditional ML: *“Is this tumor malignant or benign?”* (classification)
- ▶ Generative AI: *“Write me a Python script to parse this VCF file”* (generation)
- ▶ Generative AI: *“Explain what a p-value means to a non-statistician”* (synthesis)

Large Language Models (LLMs) are the engine behind most generative AI text tools.

They are trained on massive text corpora and learn statistical patterns in language.

How LLMs Work: The Core Idea

Your prompt → [Tokenization] → [Transformer model] → [Token prediction] → Response

Key concepts — just enough to be useful:

- ▶ **Tokens** — chunks of text ($\sim\frac{3}{4}$ of a word on average). LLMs read and write in tokens, not words.
- ▶ **Transformer architecture** — the neural network design that powers nearly all modern LLMs (GPT, Claude, Gemini, etc.).
- ▶ **Context window** — how much text the model can “see” at once. Larger windows are more useful for long documents.
- ▶ **Context** - This is the combination of prompts, attachments, response history that helps AI predict the next response. Keep context short to:
 - ▶ Reduce token consumption
 - ▶ Prevent hallucination
- ▶ **Temperature** — controls randomness. Low = deterministic/factual. High = creative/varied.
- ▶ **Training vs. inference** — training teaches the model; inference is when you use it.
- ▶ **Hallucination** - AI responses are predicted and not deterministic. Thus, they can hallucinate (ie. generate false or made up information).
- ▶ **Knowledge cutoff** - “is the point in time beyond which a large language model has not been trained on new data” – [Wikipedia](#)

Are All LLMs the Same?

No. Models differ in:

Dimension	What varies
Training data	What text the model learned from
Context window	How much input it can handle
Capabilities	Coding, reasoning, vision, tool use
Safety tuning	How the model handles sensitive requests
Access model	Free tier, API, enterprise license
Update cadence	Some models are frozen; others are updated



Tip

For bioinformatics work, differences in coding ability, context length, and access to real-time data often matter more than raw benchmark scores.

AI Tools at NIH

Generative AI Tools Available to NIH Scientists

! Important

Always check current NIH/NCI policy before sharing sensitive data with any AI tool.
Data governance requirements vary by tool and are subject to change.

- ▶ **CHIRP** — NIH's internal AI platform
- ▶ **Microsoft Copilot** — Available through NIH's Microsoft enterprise agreement
- ▶ **ChatGPT Enterprise** — OpenAI's interface
- ▶ **Claude for Government** — Anthropic's interface
- ▶ **Google Gemini for Government** — Google's interface
- ▶ **GitHub Copilot / OpenAI Codex** — Code-focused tools

Resources for Keeping Up to Date with NIH Generative AI

Websites

- ▶ NIH AI Hub
- ▶ NIH Library Guide for Generative AI use
- ▶ CBIIT Generative AI Hub
- ▶ Frederick National Lab AI Hub
- ▶ AI at HHS

Communities

- ▶ NIH GenAI Coop Teams channel
- ▶ Artificial Intelligences Interest Group
- ▶ CHIRP Teams channel

Tool Deep Dive: CHIRP

What it is: NIH's internally hosted AI assistant, designed to comply with NIH data governance requirements.

Best for:

- ▶ Tasks involving NIH-internal data
- ▶ When data residency and privacy are primary concerns
- ▶ Workflows that require NIH-sanctioned tooling

Limitations:

- ▶ May lag behind frontier models in capability
- ▶ Feature set may be more restricted than commercial tools
- ▶ Check current documentation for supported use cases
- ▶ Only connected to the NIH intranet (ie. no web searching capability)
- ▶ Work with de-identified data, pre-decision and draft policies, unpublished data, and draft manuscripts. Personally identifying information are redacted if included in input.

Note

CHIRP is an NIH-hosted option for approved sensitive-information use cases. It does not make every data type permissible.

Accessing CHIRP

To access CHIRP, goto <https://chirp.od.nih.gov>. Current models available for CHIRP include:

GPT

- ▶ Writing code
- ▶ Agentic coding
- ▶ Data analysis
- ▶ Creating documents and spreadsheets

o4-mini

- ▶ Light weight for fast reasoning, coding and visualizations tasks
- ▶ For small and less complex tasks, use o4-mini

Claude Sonnet

- ▶ Summarizing documents,
- ▶ Analyzing data
- ▶ Writing code.

GPT Image version 2

- ▶ Used for image generation

Tool Deep Dive: ChatGPT

Company: OpenAI | **Models and features:** GPT 5.6 Sol, GPT 5.5 GPT-5 Pro, GPT-5 Thinking, Deep Research, Codex (coding agent), GPT Agents, ImageGen.

To access ChatGPT, goto <https://go.hhs.gov/chatgpt>

Best for:

- ▶ General-purpose writing, summarization, Q&A
- ▶ Code generation and debugging (Python, R, Bash)
- ▶ Broad research and literature synthesis
- ▶ Iterative conversation and brainstorming

Limitations:

- ▶ 9000 credit per user limit covering ChatGPT enterprise and Codex (see [NIH Guideline KB0024005](#))
 - ▶ Users will see a pop up notification when the limit is reached
- ▶ Protected Health Information (PHI) is not allowed to be shared with ChatGPT.
- ▶ The NIH ChatGPT deployment allows for interfacing through web browser only. No access via desktop or mobile applications. (see [NIH AI Hub](#)).
- ▶ See [Data Usage Restrictions](#) for details on what data can be used in ChatGPT.

Free vs. Plus vs. Enterprise — capabilities differ significantly by tier.

Codex

Company: OpenAI | Models:** GPT Sol 5.6, GPT Terra 5.6, GPT Luna 5.6, GPT 5.5

Best for:

- ▶ Writing, editing, and explaining code
- ▶ Perform Git versioning
- ▶ Generating standup summary and pull request reviews
- ▶ Help with software implementation and deployment
- ▶ Codex is geared towards assisting with software development efforts
- ▶ API and command line access
- ▶ IDE integration (ie. VS Code)

Limitations:

- ▶ Codex is an agentic AI tool, which has full autonomy can lead to overwriting files and stealing data so be careful.
- ▶ 9000 credit per user limit covering ChatGPT enterprise and Codex (see [NIH Guideline KB0024005](#))
 - ▶ Users will see a pop up notification when the limit is reached
- ▶ Do not use:
 - ▶ Protected Health Information (PHI)
 - ▶ Sensitive Personally Identifiable Information (SPII)
 - ▶ Grant sensitive data
 - ▶ Source: <https://nih.sharepoint.com/sites/NCI-Fred/Knowledgebase/AI/SitePages/New-coding-feature-available-in-ChatGPT-Enterprise-Codex.aspx> for details on what data can be used in ChatGPT.

Tool Deep Dive: Claude

Company: Anthropic | **Models:** Claude Sonnet 5, Claude Sonnet 4.6, Claude Opus 4.8, Claude Opus 4.7, Claude Haiku 4.5

To access Claude, goto <https://claude.hhs.gov/>

Best for:

- ▶ Long documents: large context window handles entire papers or codebases
- ▶ Code writing, review, and explanation
- ▶ Step-by-step reasoning tasks
- ▶ Careful, nuanced writing (methods sections, grant language)
- ▶ Supports FedRAMP high workloads

Limitations:

- ▶ Do not use on mobile application
- ▶ Does not generate photos or images but can generate HTML/SVG based diagrams
- ▶ No Slack, Teams, SharePoint integration
- ▶ No API or command line access for NIH deployment
- ▶ See [Data Usage Restrictions](#) for details on what data can be used in Claude.



Tip

Model tiers: Haiku (fast/cheap) → Sonnet (balanced) → Opus (most capable). Same company, same product, different engines.

Tool Deep Dive: Gemini

Company: Google DeepMind | **Models and features:** Gemini pro 2.5, pro 3.1, flash 3.5, flash 3.6, flash 3.7

To access Gemini goto <https://go.hhs.gov/gemini>

Best for:

- ▶ Integration with Google Workspace (Docs, Sheets, Drive)
- ▶ Multimodal tasks (text, image, document, video as input, outputs text and image)

Limitations:

- ▶ Quality can vary by task compared to other frontier models
- ▶ Data privacy: review Google's enterprise data policy
- ▶ Integration depth varies by institution's Google Workspace agreement
- ▶ See [Data Usage Restrictions](#) for details on what data can be used in Gemini

Tool Deep Dive: Microsoft Copilot

Company: Microsoft (powered by OpenAI models)

Access at NIH: Via Microsoft 365 enterprise agreement

Access Microsoft Copilot at NIH via <https://m365.cloud.microsoft/chat>

Best for:

- ▶ Drafting and editing in Word, Outlook, PowerPoint
- ▶ Data analysis and formula help in Excel
- ▶ Integration with Teams and SharePoint (summarize Teams chats and meetings)
- ▶ Searching across your organization's documents

Limitations:

- ▶ Most powerful when used *within* Microsoft 365 apps
- ▶ Less suited for standalone bioinformatics coding tasks
- ▶ Capabilities depend on your institution's license tier

Key Interaction Principle

These are interaction modes, not product buckets.

MS Copilot is integrated into Microsoft 365 apps like Word, Excel, PowerPoint, and Teams to help with office productivity needs.

Tool Deep Dive: GitHub Copilot

Company: GitHub (Microsoft) / OpenAI

Optimized for: Code — specifically writing, completing, and explaining code

Best for:

- ▶ Inline code completion in your IDE (VS Code, JetBrains)
- ▶ Generating boilerplate code from comments
- ▶ Explaining unfamiliar code
- ▶ Writing tests and documentation
- ▶ Agentic coding
- ▶ Perform Git versioning

Limitations:

- ▶ Primarily a coding tool — not designed for general Q&A
- ▶ Suggestions can be subtly wrong; always review output
- ▶ Requires GitHub account; institutional access may vary ([LINK TO HOW TO GET ACCESS AT NIH?](#))



Tip

Copilot works *inside your editor*, not in a chat window — this makes it ideal for active development workflows.

Data Usage Restrictions

Data types: what's permitted in which tool

This table is drawn from the NIH "Responsible AI" guidance and reflects **the tool's security documentation**, not necessarily every legal/contractual restriction that might additionally apply (e.g., Data Use Agreements, CRADAs, consent documents — always check those separately with your IC's privacy/security staff).

Data Type	HHS Enterprise ChatGPT	HHS Enterprise Gemini	HHS Enterprise Claude	CHIRP Enterprise (VPN required)	M365 Copilot
Personally Identifiable Information (PII)	✔ Yes, With Restrictions*	✔ Yes, With Restrictions*	✔ Yes, With Restrictions*	✔ Yes, With Restrictions*	✔ Yes, With Restrictions*
Sensitive PII (SPII)	⊘ No	✔ Yes, With Restrictions*	✔ Yes, With Restrictions*	✔ Yes, With Restrictions*	✔ Yes, With Restrictions*
Sensitive Information	⊘ No	✔ Yes, With Restrictions*	✔ Yes, With Restrictions*	✔ Yes, With Restrictions*	✔ Yes, With Restrictions*
Individually Identifiable Health Information (IIHI)	⊘ No	⊘ No	⊘ No	⊘ No	⊘ No
Grant Sensitive Data	⊘ No	✔ Yes, With Restrictions*	✔ Yes, With Restrictions*	✔ Yes, With Restrictions*	✔ Yes, With Restrictions*
NIH Peer Review Final Products	⊘ No	✔ Yes	✔ Yes	✔ Yes	✔ Yes
Classified Information	⊘ No	⊘ No	⊘ No	⊘ No	⊘ No
Contractor Bid/Proposal & Source Selection Information	⊘ No	✔ Yes	✔ Yes	✔ Yes	✔ Yes
Information, Data, or Code Subject to Export Controls	⊘ No	⊘ No	⊘ No	⊘ No	⊘ No
SBIR/STTR Information	🔍 Under Review	🔍 Under Review	🔍 Under Review	🔍 Under Review	🔍 Under Review
Information That Could Compromise Biosafety/Biosecurity	⊘ No	⊘ No	⊘ No	⊘ No	⊘ No
Irradiator Program / Irradiator-Related Records	⊘ No	⊘ No	⊘ No	⊘ No	⊘ No
Select Agent-Related Records (42 USC 262a)	⊘ No	⊘ No	⊘ No	⊘ No	⊘ No
Security-Sensitive Operational Information	⊘ No	⊘ No	⊘ No	⊘ No	⊘ No
Sensitive Investigatory, Audit, or Compliance Information	⊘ No	⊘ No	⊘ No	⊘ No	⊘ No
Identifiable Sensitive Info Protected by Certificate of Confidentiality	⊘ No	⊘ No	⊘ No	⊘ No	⊘ No
NIH Controlled-Access Data Repositories	⊘ No	⊘ No	⊘ No	⊘ No	⊘ No

*"Yes, With Restrictions" means the tool's security documentation does not prohibit this data type — but you must still review the rules that apply to you and work with your IC's designated SMEs to confirm there are no legal or procedural restrictions for your specific use case (e.g., Data Use Agreements, CRADAs, paywalled-literature licensing terms, or research-informed consent documents). For Grant Sensitive Data questions specifically, contact OperaGrantsAI@mail.nih.gov.

Note that **HHS Enterprise ChatGPT is the most restrictive** of the five tools — it's approved only for routine non-sensitive PII (names, work emails) and already-public contract/grant materials, per HHS's own tech.hhs.gov guidance, which separately warns not to enter SPII, PHI, classified, export-controlled, or confidential commercial/trade-secret data into it.

None of these tools are the system of record for any data — authorization and consent responsibilities remain with your Institute/Center/Office, not the AI tool itself.

Source: [Responsible AI at NIH](#)

Chat vs. Agentic AI

Two Modes of Working with AI



Chat Mode

- ▶ Back-and-forth conversation
- ▶ You stay in the loop each turn
- ▶ Low setup, immediate use
- ▶ Best for: Q&A, drafting, explanation, brainstorming



Agentic Mode

- ▶ The system takes multiple actions toward a goal
- ▶ May read files, run tools, or make changes
- ▶ Requires setup, guardrails, and oversight
- ▶ Best for: multi-step tasks, automation, iterative execution

Note

These are interaction modes, not product buckets.

The same product may support one or both modes, depending on the interface and features available in your environment.

Chat Mode: What to Expect

How it works: You type a prompt → the model responds → you refine. Each exchange is called a **turn**.

Great for bioinformatics:

- ▶ Explaining commands and outputs -> “Explain what `samtools flagstat` output means”
- ▶ Brainstorming or aiding an analysis approach -> “Help me write a regex to parse this FASTQ header format”
- ▶ Comparing Methods -> “What’s the difference between DESeq2 and edgeR?”
- ▶ Reviewing or summarizing text -> “Review this methods section for clarity”

Limitations:

- ▶ Usually requires you to drive each step
- ▶ May not retain cross-session memory unless enabled
- ▶ Needs explicit tool support for files or code execution
- ▶ Outputs still need verification, especially biological claims (true of any AI)

Agentic Mode: Power and Responsibility

How it works: You define a goal → the system plans and executes multiple steps autonomously → results are returned.

- ▶ **More powerful:** Can write code, run it, read output, iterate, and produce a final result
- ▶ **More complex:** Requires version control, sandboxed environments, clear task definitions
- ▶ **More oversight needed:** Agents can make cascading mistakes without human checkpoints
- ▶ **More expertise required:** Setup, debugging, and trust calibration are non-trivial

Warning

Best practices for agentic tools:

- ▶ Use version control.
- ▶ Review outputs at checkpoints.
- ▶ Define scope clearly before execution.
- ▶ Start with small, reversible tasks before scaling up.

Guiding Agentic AI: Project Instructions and Skills

Project instructions

- ▶ Describe the current project or repository
- ▶ In many tools this appears as AGENTS.md
- ▶ Tell the agent what this project is, where the files are, and what not to change
- ▶ Sets guardrails and expectations for the local context

Skills

- ▶ Describe a reusable procedure for one specific task
- ▶ Usually packaged as a skill folder with a SKILL.md file
- ▶ Includes YAML front matter
- ▶ Give the agent step-by-step instructions for a repeatable workflow

Note

Project instructions are like a README for the agent. They explain the project context, conventions, and guardrails. Skills tell the agent how to do the work.

Agentic vs. Chat: Choosing the Right Mode

Scenario	Recommended Mode
"Explain this error message"	 Chat
"Help me write a DESeq2 script"	 Chat
"Build and run a QC pipeline on my FASTQ files"	 Agentic
"Refactor all scripts in this repo"	 Agentic
"Summarize this paper for me"	 Chat
"Monitor a workflow and resubmit failed jobs"	 Agentic

Rule of thumb: Start with chat. Graduate to agentic when the task is well-defined, repeatable, and you understand what the agent should do at each step.

AI in Bioinformatics Practice

What Can AI Actually Help With?

Learning & Education

- ▶ Explain a concept at the right level (“explain PCA for someone who knows basic statistics”)
- ▶ Summarize and compare methods papers
- ▶ Generate study questions from a paper
- ▶ Walk through an error message step by step

What Can AI Actually Help With?

Coding & Pipeline Development

```
1 # Prompt: "Write a Python script to parse a GTF file and  
2 #           return a dictionary of gene_id to transcript_id"
```

- ▶ Write boilerplate code for common file formats (FASTQ, VCF, BED, GTF, SAM)
- ▶ Debug error messages — paste the traceback, get an explanation
- ▶ Convert code between languages (R → Python, Python → Bash)
- ▶ Explain what someone else's code does
- ▶ Add comments and documentation to existing scripts

What Can AI Actually Help With?

Workflow & Analysis Support

- ▶ Suggest appropriate statistical tests for a given experimental design
- ▶ Help design a bioinformatics pipeline (what tools, what order, what parameters)
- ▶ Draft a methods section based on the tools and parameters you used
- ▶ Generate shell commands for file manipulation, renaming, or batch processing
- ▶ Improve visualizations: “make this ggplot2 figure color-blind friendly”

What Are the Limitations?

 Know These Before You Trust the Output

- ▶ **Hallucination** — LLMs generate plausible-sounding text, including plausible-sounding *wrong* text. Gene names, database IDs, citations, and parameter recommendations can be fabricated.
- ▶ **Knowledge cutoff** — models have a training date. Recent tools, updated databases, and new methods may be unknown to the model.
- ▶ **No real-time access** (unless explicitly enabled) — the model cannot check PubMed, Ensembl, or NCBI live.
- ▶ **Context limits** — very long inputs may be truncated or cause degraded performance.
- ▶ **No self-awareness of error** — models don't know what they don't know. They will answer with confidence regardless.

The Verification Imperative

Always verify:

- ▶ Any specific biological claim (pathway, gene function, interaction)
- ▶ Database IDs, accession numbers, citations
- ▶ Code logic — test on small examples before scaling
- ▶ Statistical recommendations — check against the methods literature
- ▶ Tool parameters — consult the tool's documentation



Tip

A useful mindset:

Treat AI output the way you'd treat advice from a very smart first-year graduate student.

Smart, fast, helpful — but always check their work.

Prompt Engineering: Getting Better Results

Small changes in how you ask make a big difference.

► **Basic** ► Better ► Best

Explain STAR alignment.

Getting Started

Know Your Experience Level

A Critical Point Before We Go Further

AI can generate code that *looks* correct. In biology, looking correct and *being* correct are not the same thing. **Publishing results from unvalidated AI-generated code is a real and growing risk.**

Your computational experience matters more than your job title. What matters most is your ability to recognize when AI output is wrong.

- ▶ **Less computational experience?** Use AI to *understand* — learn concepts, interpret outputs someone else has validated, draft text, summarize literature, and ask “does this result make sense?”
- ▶ **More computational experience?** Use AI to *accelerate* — write and debug code faster, explore unfamiliar tools or languages, generate starting-point pipelines, draft documentation.
- ▶ **Everyone, regardless of experience:** Always review AI-generated logic. Test on known datasets before applying to real data. Never assume AI-chosen parameters fit your experimental design without checking.

Important

-  Do NOT run AI-generated code on your data and publish it without someone who has the expertise to validate it, whether that's you or a collaborator.
-  **You are responsible for the output, not the AI.** This holds true no matter where you fall on the spectrum.

The “Plug and Chug” Problem

⚠ A Hypothetical Scenario.

A wet lab biologist generates RNA-seq data. They paste their file paths into ChatGPT and ask for a differential expression analysis script. ChatGPT produces 80 lines of clean, well-commented R code. They run it. It finishes without errors. They publish the results.

What could go wrong?

- ▶ Low-count pre-filtering thresholds were not appropriate for the library depth
- ▶ The model matrix did not correctly encode the experimental design
- ▶ Batch effects were not accounted for — the AI didn't know about them
- ▶ The AI used a deprecated function that silently returns wrong results in the current package version
- ▶ **No errors were thrown. The figures looked reasonable. The biology was wrong.**



Tip

The solution is not to avoid AI. It is to ensure the person using it has, or works with someone who has, the expertise to validate the output.

Foundational Knowledge Is Not Optional

AI tools are **amplifiers** — they strengthen what you already know and can help you build new knowledge faster.

You bring...	AI can help with...
Understanding of your experimental design	Translating that design into code
Knowledge of appropriate statistical methods	Implementing those methods correctly
Familiarity with the data format and QC expectations	Writing parsers and QC checks
Domain expertise to interpret results	Explaining what the output means
Awareness of common failure modes	Catching when the output doesn't make sense

⚠ Important

If you don't have the foundation to recognize a wrong answer, you cannot safely use AI to generate answers.

This is not a criticism. It's a call to invest in training. That's why this series exists.

Your First Week with AI Tools

1. **Pick one tool and stick with it initially** — don't try to evaluate five at once
2. **Start with low-stakes tasks** — explaining concepts, cleaning up prose, simple code
3. **Keep a “prompt journal”** — note what worked and what didn't
4. **Be specific in your prompts** — vague questions get vague answers
5. **Iterate, don't accept the first response** — ask follow-ups, push back, ask for alternatives
6. **Always verify outputs** before using them in your work

Training Resources

Upcoming Trainings:

- ▶ NIH AI Hub: <https://nih.sharepoint.com/sites/NIH-ai/SitePages/NIH-AI-Training-Resources.aspx>
 - ▶ Topics include:
 - ▶ AI literacy
 - ▶ How to use specific tools
 - ▶ Best Practices
- ▶ BTEP: <https://bioinformatics.ccr.cancer.gov/btep/>

Presentation Recordings are available from the NIH Library and NIH GenAI CoP:

- ▶ NIH Library: <https://www.nihlibrary.nih.gov/training/recordings-ai-series>
- ▶ NIH GenAI CoP: [SharePoint Site](#)
- ▶ BTEP: <https://bioinformatics.ccr.cancer.gov/btep/btep-video-archive-of-past-classes/>

What's Coming in This Series

Session	Title	Focus
1	Introduction (<i>today</i>)	Concepts, tools, roles, limitations
2	From Scripts to Skills: Using Generative AI Without Losing Scientific Control	Combining domain knowledge, explicit context, executable checks, reusable instructions, and human judgment
3	Responsible AI Use, Validation, and Reproducibility in Bioinformatics	Reproducibility, documentation, and validation best practices
4	Integrating Generative AI into Robust Bioinformatic Analyses	End-to-end bioinformatics case study

Questions?

Thank you!

NCI CCR Bioinformatics Training and Education Program

Generative AI in Bioinformatics Seminar Series — Session 1

Contact Us: ncibtep@nih.gov

“AI is an amplifier. Bring something worth amplifying.”

These materials are designed to be updated as tools and policies evolve.

Last reviewed: 2026-09-23